

Supporting Information

Oxygen-Vacancy Energetics and Dopant-Induced Holes in Pr- and Rb-Substituted Anatase TiO₂: A DFT+U Comparison

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Revision note (v4; delete before submission). Section numbering, table numbering and figure numbering follow the v4 main text. Items marked **[AUTHOR ACTION]** need data that only the authors can extract from the calculation records. Tables S9 and S10 are new and are the single traceable account of what was calculated that the audit asked for.

Contents. S1 Pristine V_O convergence · S2 Additional PDOS · S3 Hubbard-U sweep · S4 k-mesh verification · S5 Bader analysis · S6 Alternative V_O sites · S7 Hybrid, dual-U and dispersion calculations · S8 Methods details and file inventory · S9 Full-precision energy ledger · S10 Run manifest · S11 Force-convergence trajectories

S1. Pristine V_O baseline: convergence recovery

The pristine anatase V_O reference reported in Table 2 of the main text ($E_f = +4.74$ eV) required two attempts. The first used `mixing_mode = local-TF` with `mixing_beta = 0.3`, not the damped `mixing_beta = 0.1` of the doped-cell relaxations (Section 2.6), and oscillated, with the SCF accuracy estimate moving between 0.05 and 0.14 Ry over iterations 14–17, an instability associated with the two vacancy electrons occupying a dense manifold of conduction-band states. The second attempt (run `Pristine_VO_v3`) changed six settings together: `mixing_beta` 0.3 → 0.1; `mixing_ndim` → 12; Gaussian smearing `degauss` 0.005 → 0.01 Ry (≈ 136 meV); `electron_maxstep` → 300; `diago_thr_init` → 1×10^{-3} Ry; and `starting_magnetization` = 0.05 on both species as a symmetry-breaking nudge. The SCF then converged monotonically in 96 iterations of the first BFGS step to `conv_thr` = 1×10^{-8} Ry. Six further BFGS steps brought the maximum force to 0.00563 Ry/Bohr, slightly above the 0.005 Ry/Bohr production threshold, at which point the relaxation was stopped. The final total energy of the 47-atom cell is -4233.15036 Ry; against $E(\text{Pristine_perfect}) = -4275.01530$ Ry and $\frac{1}{2}E(\text{O}_2) = -41.51621$ Ry this gives $E_f = +4.74$ eV. The converged cell is non-magnetic. **[AUTHOR ACTION: state whether the production doped-cell relaxations used `degauss = 0.005` or `0.01` Ry; if the pristine V_O cell used a different smearing from the doped cells, say so in Section 2.4 of the main text.]**

S2. Additional projected densities of states

S2.1 Pr_VO with dopant channels. Figure S1 shows the PDOS of the Pr_VO cell with the Pr-5d and Pr-5p channels plotted separately from Ti-3d and O-2p. The conduction band is dominated by Ti-3d; the Fermi level lies inside the conduction band because the cell carries one net excess electron (Table 1 of the main text). The Pr.pbe-spdn-kjpaw_psl.1.0.0 dataset has $Z_{\text{val}} = 11$ ($5s^25p^65d^1(6s,6p)^2$, Table 1 of the main text) with the 4f electrons frozen in the core, so no 4f projection exists. **[AUTHOR ACTION: the interim artwork has the empty channel's legend entry and the interpretive text box patched out; regenerate Figure S1 from projwfc.x output with channels Ti-3d, O-2p, Pr-5d, Pr-5p and total only and state the Gaussian broadening.]** The small residual between the total DOS and the sum of the projected channels arises from the incompleteness of the atomic projectors, not from a missing 4f channel.

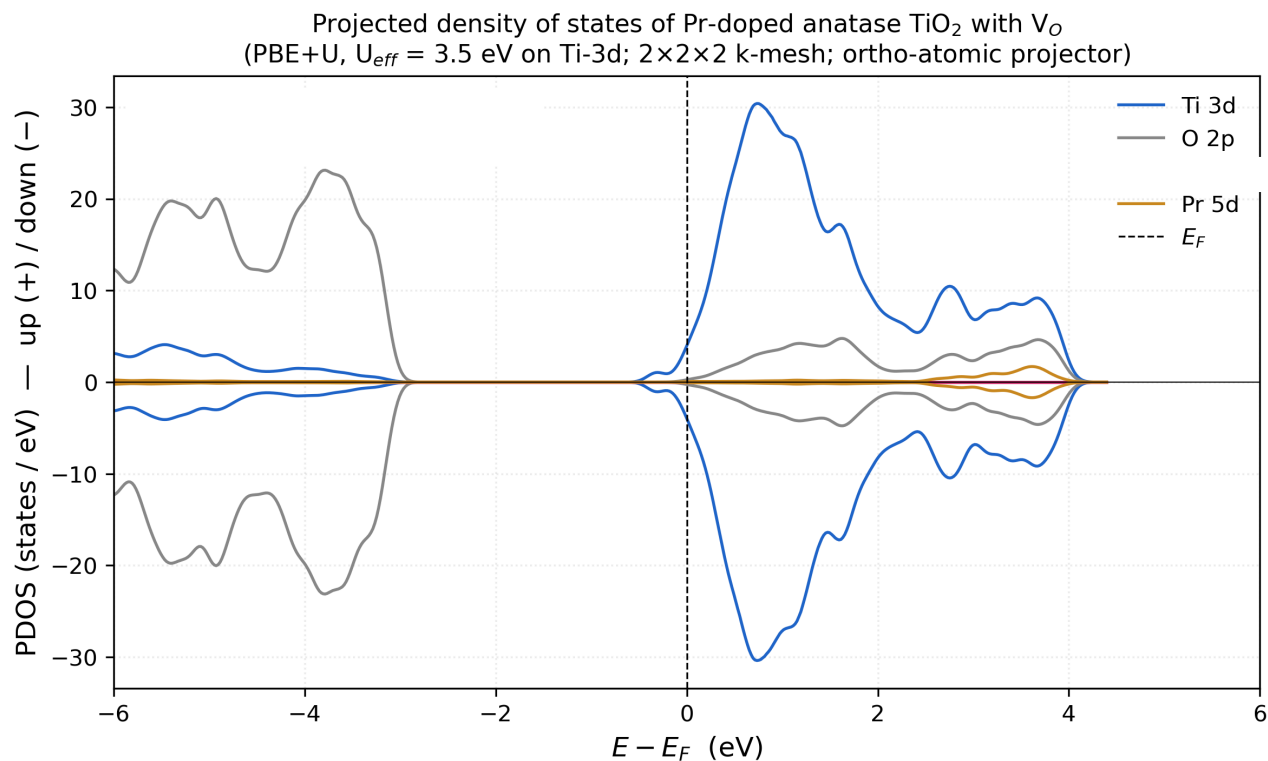


Figure S1. Projected density of states of the Pr_VO cell (PBE+U, $U(\text{Ti-3d}) = 3.5$ eV, $2 \times 2 \times 2$ non-self-consistent, $E_F = 10.306$ eV). Channels: Ti-3d summed over 15 Ti; O-2p summed over 31 O; Pr-5d; Pr-5p; total. **[AUTHOR ACTION: artwork to be regenerated as noted in Section S2.1; delete this note once done.]**

S2.2 Total DOS, Pr_VO versus Rb_VO. Figure S2 shows the non-spin-resolved companion of Figure 4 of the main text. Pr_VO: $E_F = 10.31$ eV inside the Ti-3d conduction band. Rb_VO: $E_F = 7.14$ eV at the top of the O-2p valence band, with a small feature straddling E_F and centred just above it. **[AUTHOR ACTION: the panel subtitles of the interim artwork were patched (“singlet” removed; m_{abs} instead of $|m|$); regenerate from the projwfc.x output.]**

Projected DOS: Pr-doped vs Rb-doped anatase TiO₂ with V_O

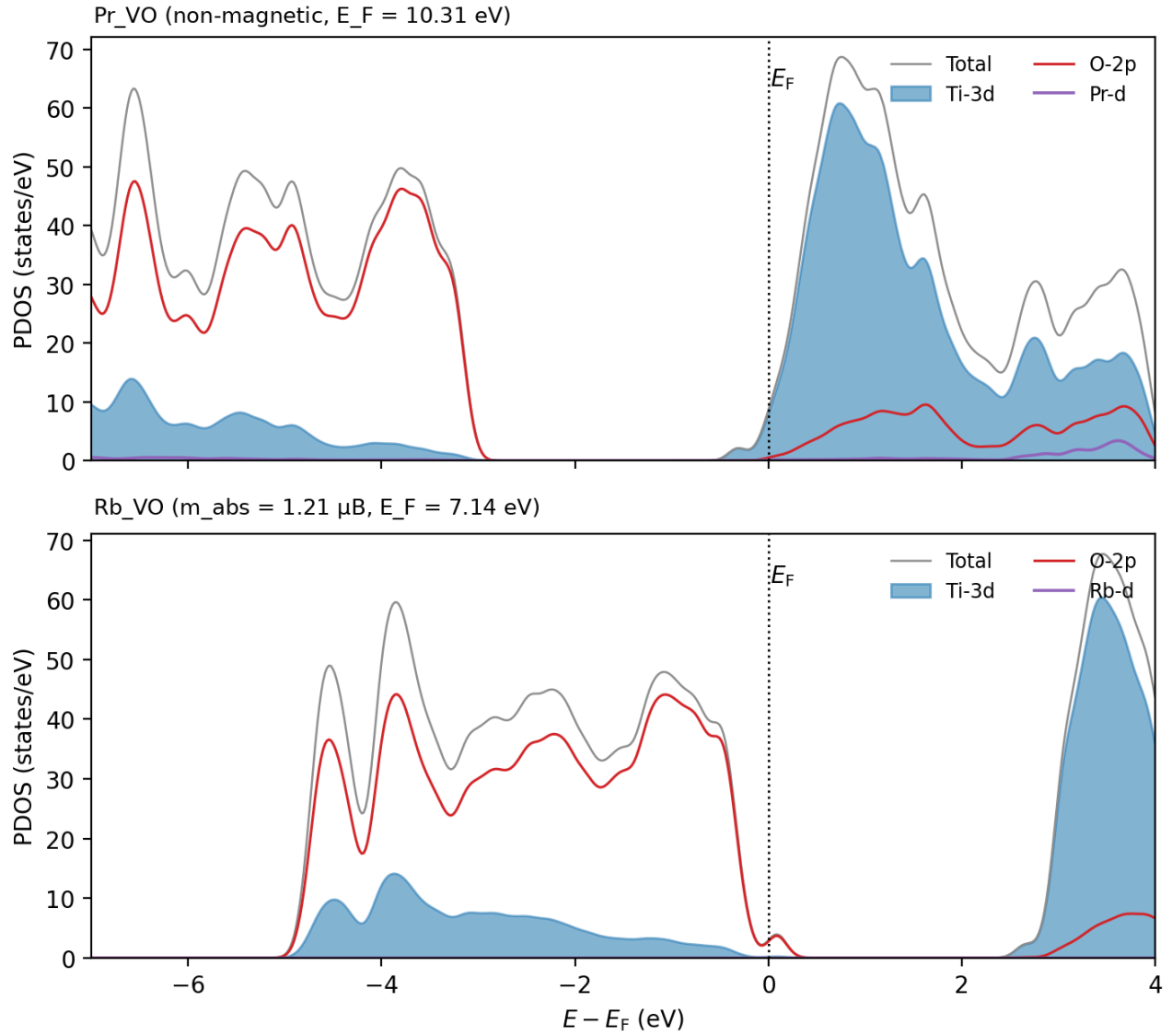


Figure S2. Projected density of states of Pr_VO (top) and Rb_VO (bottom) without spin decomposition. Filled blue: Ti-3d; red: O-2p; purple: dopant d; grey: total.

S3. Hubbard-U sensitivity sweep

Single-point calculations on the relaxed Pristine_perfect, Pr_VO and Rb_VO cells at $U(\text{Ti-3d}) = 3.0, 4.0$ and 4.5 eV (Γ -only, 40 Ry, ortho-atomic projectors). The maximum variation of the printed Fermi energy across the sweep is 0.26 eV.

Table S1. Absolute magnetisation m_{abs} (μ_B) across the $U(\text{Ti-3d})$ sweep. The production (3.5 eV) column is from the relaxed calculation; the others are single points on the same geometry.

Cell	U = 3.0	U = 3.5 (production)	U = 4.0	U = 4.5
Pristine_perfect	0.00	0.00	0.00	0.00
Pr_VO	0.00	0.00	0.00	0.00
Rb_VO	1.00	1.21	1.00	1.00

[AUTHOR ACTION: add the total energies and Fermi energies of the nine single points as additional columns, with run identifiers.]

S4. k-mesh verification

Single-point SCF energies on the Γ -relaxed geometries at a $2 \times 2 \times 2$ Monkhorst-Pack mesh (40 Ry, $U(\text{Ti-3d}) = 3.5$ eV).

Table S2. Total energies at Γ -only and $2 \times 2 \times 2$ (Ry) and the resulting formation energies (eV).

Cell	E_{tot}, Γ (Ry)	$E_{\text{tot}}, 2 \times 2 \times 2$ (Ry)	$\Delta E = E(2 \times 2 \times 2) - E(\Gamma)$ (Ry)	ΔE (eV)
Pr_perfect	-4576.71224	-4576.62183	+0.09041	+1.230
Pr_VO	-4535.10106	-4535.01814	+0.08292	+1.128
Rb_perfect	-4330.38890	-4330.28761	+0.10129	+1.378
Rb_VO	-4289.17135	-4289.08034	+0.09101	+1.238
$E_f(\text{V}_\text{O}; \text{Pr})$	+1.292 eV	+1.190 eV		
$E_f(\text{V}_\text{O}; \text{Rb})$	-4.063 eV	-4.203 eV		
$\Delta E_f(\text{Pr} - \text{Rb})$	+5.356 eV	+5.394 eV		$\Delta \Delta E = +0.038$ eV

$\Delta \Delta E$ is defined as the $2 \times 2 \times 2$ contrast minus the Γ -only contrast. The main text (Section 3.7) also quotes energies for Pristine_perfect, Pr_VO and Rb_VO at $2 \times 2 \times 1$, $2 \times 2 \times 2$ and $3 \times 3 \times 2$ meshes (magnitudes of $E(\Gamma) - E(3 \times 3 \times 2)$: 1.05, 1.08, 1.16 eV; of $E(2 \times 2 \times 1) - E(3 \times 3 \times 2)$: 1.15, 1.05, 0.42 meV/atom). In the table above $\Delta E = E_{\text{tot}}(2 \times 2 \times 2) - E_{\text{tot}}(\Gamma)$ is positive, so the $2 \times 2 \times 2$ energies lie above the Γ -only ones; they agree with the quoted $3 \times 3 \times 2$ values to 0.05–0.08 eV only if the Γ -only energies lie below $3 \times 3 \times 2$. **[AUTHOR ACTION: state the sign of $E(\Gamma) - E(3 \times 3 \times 2)$ for each cell from the output files.]** **[AUTHOR ACTION: add all of those energies to this table with run identifiers, including Pristine_perfect at $2 \times 2 \times 2$; the pristine V_O cell was not computed at $2 \times 2 \times 2$. The earlier run log listed only Pr_perfect, Rb_perfect and Pr_VO as completed for the $2 \times 2 \times 2$ stage although an Rb_VO value is reported above; confirm the Rb_VO $2 \times 2 \times 2$ run from its output file.]**

S5. Bader topological-charge analysis

Bader analysis of the relaxed Pr_VO and Rb_VO cells: pp.x charge density (plot_num = 0, output_format = 6, Gaussian cube) decomposed with the Henkelman grid-based code (v1.05, 2023-08-19). **[AUTHOR ACTION: state the FFT grid dimensions and whether the all-electron PAW density was reconstructed; if not, say that the comparison uses the smooth valence density.]**

Table S3. Summary Bader metrics. Charges are Bader electron populations in e; “depletion” is the population difference $\text{Rb_VO} - \text{Pr_VO}$ for the corresponding atom.

Quantity	Pr_VO	Rb_VO
Total electrons (Bader sum)	376.998	374.998
Mean Ti population (15 atoms)	9.683	9.671
Mean O population (31 atoms)	7.187	7.155
Dopant Bader charge (Z_{val} minus population)	Pr: +2.14 e	Rb: +0.82 e

Quantity	Pr_VO	Rb_VO
O sublattice total difference (Rb_VO – Pr_VO)	reference	–0.97 e
Largest single-atom O depletion	reference	–0.257 e on O8 (2nd shell of Rb)
Second-largest O depletion	reference	–0.14 e on O30
Mean Ti depletion	reference	–0.012 e per Ti (–0.18 e total); none exceeds –0.05 e

The Bader sums reproduce the valence electron counts of Table 1 of the main text (377 and 375) and thereby confirm the $Z_{\text{val}} = 11$ (4f-in-core) Pr dataset. **[AUTHOR ACTION: the dopant charges are inconsistent with the mean populations above (sum rule: $376.998 - 15 \times 9.683 - 31 \times 7.187 = 8.956$ e on Pr, i.e. +2.04 e; $374.998 - 15 \times 9.671 - 31 \times 7.155 = 8.128$ e on Rb, i.e. +0.87 e); re-extract all values from ACF.dat, add the full 47-row per-atom tables for both cells as Tables S3a and S3b, state that both cells use the same atom ordering and which oxygen index was removed, and note that grid convergence of the Bader charges was not tested. Add the spin-density isosurface of Rb_VO and the site-resolved projections of its gap state as Figure S4.]**

S6. Alternative V_O sites in the Pr cell

Table S4. V_O formation energies at four selected oxygen sites of the Pr-substituted cell (BFGS, $U(\text{Ti-3d}) = 3.5$ eV only, Γ -only, 40 Ry). Sites O1 and O17 are symmetry-equivalent; their agreement to 0.1 meV is an internal consistency check.

Site (index in production input)	Pr-O distance (Å)	E_{tot} (Ry)	$E_{\text{f}}(V_O)$ (eV)	Relative to production site (eV)
Production Pr_VO (nearest O to Pr)	≈ 2.17 (1st shell)	–4535.10106	+1.29	0
O1 (2nd shell)	3.99	–4535.04367	+2.07	+0.78
O7 (distant)	4.94	–4534.99608	+2.72	+1.43
O17 (2nd shell, equivalent to O1)	3.99	–4535.04366	+2.07	+0.78

[AUTHOR ACTION: add the Cartesian coordinates of the four removed oxygen atoms and the minimum-image Pr- V_O distances; state the index of the production vacancy.]

S7. Hybrid-functional, dual-Hubbard and dispersion calculations

Table S5. Results of the additional calculations (Sections 3.4, 3.6 and 3.7 of the main text).

Calculation	Cell	E_{tot} (Ry)	Key observable
HSE06 single point ($\alpha = 0.25$, $\omega = 0.106$ Bohr ^{–1})	Pristine_perfect	–3926.66858	$E_{\text{F}} = 8.286$ eV; $m_{\text{abs}} = 0$
Dual-U relaxation ($U(\text{Ti-3d}) = 3.5$ eV, $U(\text{O-2p}) = 5.5$ eV)	Pr_perfect	–4571.33095	$m_{\text{abs}} = 1.00 \mu_{\text{B}}$; $E_{\text{F}} = 6.69$ eV; $F_{\text{max}} = 0.0049$ Ry/Bohr
Dual-U (control)	Pristine_perfect	–4269.72111	$m_{\text{abs}} = 0$; $E_{\text{F}} = 7.22$ eV
Dual-U	Rb_perfect	[AUTHOR ACTION: E_{tot}]	$m_{\text{abs}} = 3.00 \mu_{\text{B}}$

Calculation	Cell	E_{tot} (Ry)	Key observable
Dual-U	Rb_VO	[AUTHOR ACTION: E_{tot}]	$m_{\text{abs}} = 1.00 \mu_{\text{B}}$
D3(BJ) single point (dftd3_version = 4, no three-body)	Pr_VO	-4535.85464	D3 term = -10.29 eV; $m_{\text{abs}} = 0$; $E_{\text{F}} = 9.99$ eV
D3(BJ) single point	Rb_VO	-4289.91254	D3 term = -10.09 eV; $m_{\text{abs}} = 1.00 \mu_{\text{B}}$; $E_{\text{F}} = 7.12$ eV

The D3 single points were not performed on Pr_perfect, Rb_perfect or O₂, so no D3-corrected formation energy can be formed (main text Section 3.7). The HSE06 total energy is not comparable with the PBE+U totals.

S8. Methods details and file inventory

Quantum ESPRESSO v7.5; PSLibrary 1.0.0 PAW datasets Ti.pbe-spn-kjpaw_psl.1.0.0.UPF, O.pbe-n-kjpaw_psl.1.0.0.UPF, Pr.pbe-spdn-kjpaw_psl.1.0.0.UPF ($Z_{\text{val}} = 11$; 4f in core) and Rb.pbe-spn-kjpaw_psl.1.0.0.UPF; nspin = 2; HUBBARD (ortho-atomic) with $U(\text{Ti-3d}) = 3.5$ eV and, for the dual-U runs, $U(\text{O-2p}) = 5.5$ eV. **[AUTHOR ACTION: Table S7 — paste the “Valence configuration” block and SHA-256 hash of each UPF file. Table S8 — relaxed first-shell dopant-O distances for Pr and Rb.]**

The input templates, orchestrator scripts, raw output files, ACF.dat files, PDOS files and the master spreadsheet are deposited on Zenodo (DOI in the Data availability statement of the main text). Local directory names of the authors’ computing facility are not reported.

Table S6. Wall-time summary (Intel i5-3570, 4 cores).

Stage	Wall time	Outcome
Production single-U relaxations (5 cells)	≈ 3 days each	below 0.005 Ry/Bohr (Pr_VO stopped manually; Section 2.6)
Pristine V _O relaxation (after mixing recovery, S1)	≈ 14 h	$F_{\text{max}} = 0.0056$ Ry/Bohr
HSE06 single point, Pristine_perfect	8 days	done
Dual-U relaxations, Pr_perfect and Pristine	≈ 24 h	done
Dual-U calculations (relaxation or single point to be stated, Section 3.4), Rb_perfect and Rb_VO	[AUTHOR ACTION]	done
D3 single points, Pr_VO and Rb_VO	≈ 12 h total	done
Alternative-site Pr_VO relaxations (3)	≈ 33 h total	done
Bader, Pr_VO and Rb_VO	≈ 2 h	done (Pristine not completed)
2×2×2 single points (4 doped cells; Pristine_perfect in the k-mesh series below)	≈ 6 h total	done [AUTHOR ACTION: confirm the Rb_VO run, see S4]
Charged cells (4 relaxations)	[AUTHOR ACTION]	done
U sweep (9 single points)	[AUTHOR ACTION]	done
k-mesh series at 2×2×1, 2×2×2, 3×3×2 (3 cells)	[AUTHOR ACTION]	done

S9. Full-precision energy ledger

All values are the “! total energy” lines printed by pw.x (smearing free energies for the smeared cells; see Section 2.4 of the main text); conversions use $1 \text{ Ry} = 13.605693 \text{ eV}$. In the Protocol column, single-U denotes $U(\text{Ti-3d}) = 3.5 \text{ eV}$ only and dual-U denotes $U(\text{Ti-3d}) = 3.5 \text{ eV}$ plus $U(\text{O-2p}) = 5.5 \text{ eV}$. Formation energies in the main text are computed from these values. N_e of the charged cells: Pr_VO 376 ($q = +1$) and 375 ($q = +2$); Rb_VO 374 ($q = +1$) and 373 ($q = +2$).

Table S9. Energy ledger.

Run	Cell	q	Protocol	k-mesh	E_{tot} (Ry)	Derived quantity
1	Pristine_perfect	0	single-U	Γ	-4275.01530	reference
2	Pristine_VO_v3	0	single-U	Γ	-4233.15036	$E_f = +4.745 \text{ eV}$
3	Pr_perfect	0	single-U	Γ	-4576.71224	reference
4	Pr_VO (production site)	0	single-U	Γ	-4535.10106	$E_f = +1.292 \text{ eV}$
5	Rb_perfect	0	single-U	Γ	-4330.38890	reference
6	Rb_VO	0	single-U	Γ	-4289.17135	$E_f = -4.063 \text{ eV}$
7	O ₂ (12 Å box, triplet)	0	PBE	Γ	$2 \times (-41.51621)$ [AUTHOR ACTION: paste the printed “! total energy” line of the O₂ run]	$\frac{1}{2}E(\text{O}_2) = -41.51621 \text{ Ry} = -564.857 \text{ eV}$
8	Pr_perfect	0	single-U	$2 \times 2 \times 2$	-4576.62183	
9	Pr_VO	0	single-U	$2 \times 2 \times 2$	-4535.01814	$E_f = +1.190 \text{ eV}$
10	Rb_perfect	0	single-U	$2 \times 2 \times 2$	-4330.28761	
11	Rb_VO	0	single-U	$2 \times 2 \times 2$	-4289.08034	$E_f = -4.203 \text{ eV}$
12	Pr_VO, site O1	0	single-U	Γ	-4535.04367	$E_f = +2.073 \text{ eV}$
13	Pr_VO, site O7	0	single-U	Γ	-4534.99608	$E_f = +2.720 \text{ eV}$
14	Pr_VO, site O17	0	single-U	Γ	-4535.04366	$E_f = +2.073 \text{ eV}$
15	Pr_VO	+1	single-U, uniform background	Γ	-4535.83092	spin diagnostic only
16	Pr_VO	+2	single-U, uniform background	Γ	-4536.36183	spin diagnostic only
17	Rb_VO	+1	single-U, uniform background	Γ	-4289.68663	spin diagnostic only
18	Rb_VO	+2	single-U, uniform background	Γ	-4290.19380	spin diagnostic only

Run	Cell	q	Protocol	k-mesh	E _{tot} (Ry)	Derived quantity
19	Pr_perfect	0	dual-U	Γ	-4571.33095	m _{abs} = 1.00 μ _B
20	Pristine_perfect	0	dual-U	Γ	-4269.72111	m _{abs} = 0
20a	Rb_perfect	0	dual-U	Γ	[AUTHOR ACTION]	m _{abs} = 3.00 μ _B
20b	Rb_VO	0	dual-U	Γ	[AUTHOR ACTION]	m _{abs} = 1.00 μ _B
21	Pristine_perfect	0	HSE06	Γ	-3926.66858	not comparable with PBE+U totals
22	Pr_VO	0	single-U + D3(BJ)	Γ	-4535.85464	E(D3) – E(run 4) = -10.25 eV
23	Rb_VO	0	single-U + D3(BJ)	Γ	-4289.91254	E(D3) – E(run 6) = -10.08 eV
24	Pr_VO, magnetic trial	0	single-U, magnetic start	Γ	[AUTHOR ACTION]	+0.281 eV above run 4; m = 1.00, m _{abs} = 1.13 μ _B

Derived differences: E_f(Pr) – E_f(Pristine) = -3.453 eV; E_f(Rb) – E_f(Pristine) = -8.808 eV; ΔE_f(Pr – Rb) = 5.356 eV (Γ), 5.394 eV (2×2×2); ΔΔE = +0.038 eV.

S10. Run manifest

Table S10. Run manifest (template to be completed from the output files). One row per pw.x run that contributes a number to the main text or SI.

Run ID	Output file	Cell	Composition	N _e	q	Functional / U (eV)	k-mesh	E _{cut} (Ry)	degauss (Ry)	starting_magn
1		Pristine_perfect	Ti ₁₆ O ₃₂	384	0	PBE+U 3.5	Γ	40/320		[AUTHOR ACTION: value from the not 0 on every Section 2.5]
...										

[AUTHOR ACTION: complete all 24+ rows from the output files; this table is what reviewers will use to trace every number.]

S11. Force-convergence trajectories

Force convergence of geometry-relaxed defect cells

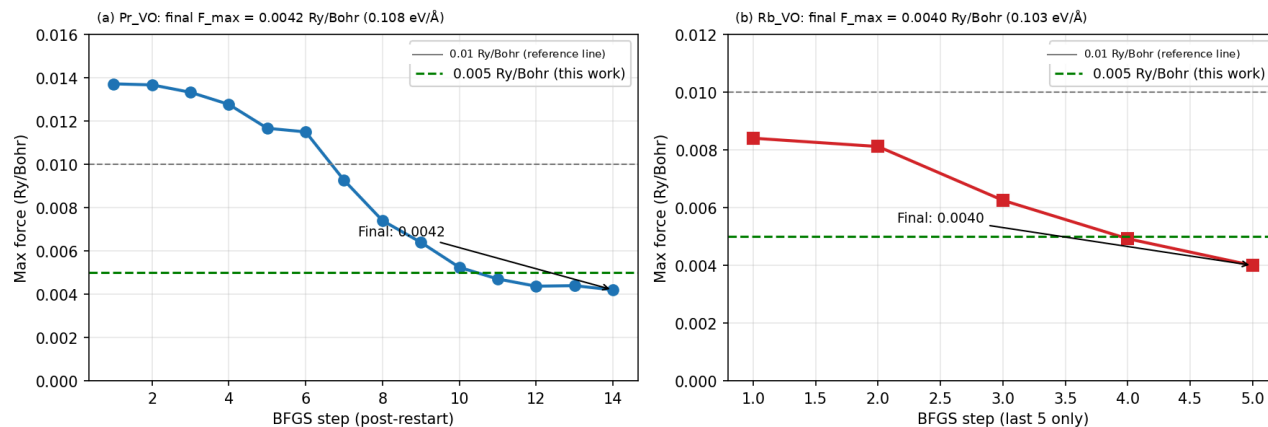


Figure S3. Maximum-force trajectories of the (a) Pr_VO (last 14 BFGS steps after restart) and (b) Rb_VO (last 5 steps) relaxations. Dashed green: the 0.005 Ry/Bohr threshold of this work. Final values 0.0042 and 0.0040 Ry/Bohr (0.108 and 0.103 eV/Å). **[AUTHOR ACTION: the panel titles and the legend entry of the grey 0.01 Ry/Bohr line (formerly labelled as a literature standard without a source) were patched in the interim artwork; regenerate from the BFGS logs with labels in eV/Å, drop or source the 0.01 Ry/Bohr line, and show the complete trajectories.]**